Cross-Model Transferability of Hallucination Signals in Large Language Models

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Abstract

Large language models can generate confident but factually incorrect responses, creating reliability concerns for real-world deployment. Many hallucination detectors rely on model-internal representations such as hidden states, activations, or latent directions, which limits their reusability across model families and API-only settings. This paper studies whether output-side signals can support hallucination detection that transfers across LLM families. Using CommonsenseQA and responses from Llama 3.1 8B, Mistral 7B, and Gemma2 9B, we represent each response with entropy, top-2 margin, length-normalized log probability, and token count. We train a lightweight MLP detector and evaluate same-model and cross-model performance. Results show that combining output-side signals outperforms single-feature baselines and that detectors retain useful performance across model families, suggesting that hallucination-related output patterns are not entirely model-specific. The codebase can be found at github.com/santiagoposse/Cross-Model-Transferability-of-Hallucination-Signals.

1. Introduction

Large language models (LLMs) can generate fluent and confident responses that are factually incorrect. This behavior creates reliability concerns in settings where users may not be able to easily distinguish correct answers from incorrect ones (Alansari & Luqman, 2026).

Prior work detects hallucinations using uncertainty, semantic variation, and generation behavior (Huang et al., 2025; Yang et al., 2023; Farquhar et al., 2024). More recent methods use model-internal representations, including hidden states, layer-wise activations, symbolic layer behavior, and latent directions associated with truthfulness (Mir, 2025; Lamba et al., 2025; Zhang et al., 2025; Park et al., 2025). These internal signals can be useful, but they limit reusability because hidden states and activations are often unavailable in API-only or closed-model settings. Detectors trained on one model family may also fail to transfer cleanly to models with different architectures, tokenizers, or training procedures, especially under distribution shift (Dasgupta et al., 2025).

This paper focuses on whether hallucination detection can be performed using only output-side signals, and whether those signals transfer across different LLM families. If hallucination-related output patterns are not entirely model-specific, then a detector trained on one model's outputs may retain useful performance on another model's outputs.

We study this approach using CommonsenseQA (Talmor et al., 2019) and three open-weight model families including Llama 3.1 8B (Grattafiori et al., 2024), Mistral 7B (Jiang et al., 2023), and Gemma2 9B (Team et al., 2024). Each generated response is labeled by answer correctness and represented using entropy, top-2 margin, length-normalized log probability, and token count. We train a lightweight multilayer perceptron (MLP) detector and evaluate it in same-model and cross-model settings.

Our results show that combining output-side signals improves over single-feature logistic regression baselines. Detectors trained on one model also retain meaningful performance on different model families, suggesting that some hallucination-related output patterns transfer across models.

2. Related Work

2.1. Hallucination and Uncertainty in LLMs

Hallucination in LLMs is a major reliability problem because generated answers may appear fluent and confident while still being factually incorrect. Survey work summarizes hallucination causes, detection methods, and mitigation strategies across model and task settings (Alansari & Luqman, 2026).

One line of work detects unreliable generations using uncertainty and output behavior. Uncertainty analysis has been used to study when LLM generations are less reliable

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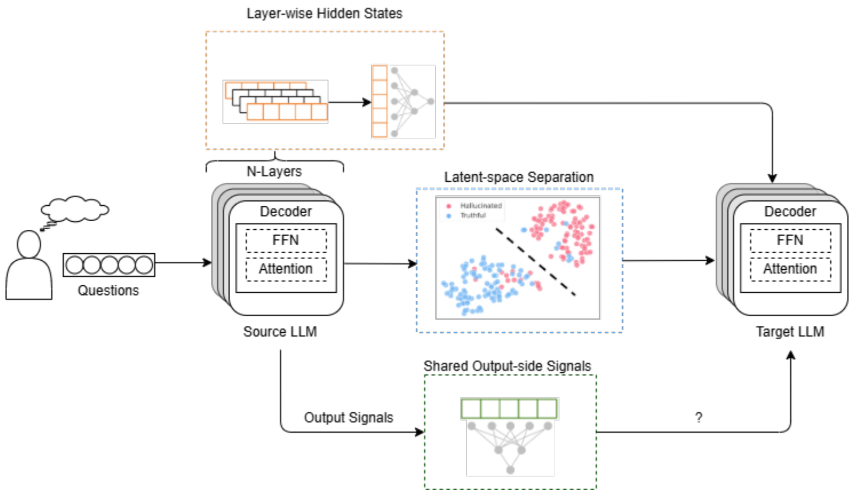

Figure 1. Gap between internal representation hallucination detectors and output-side detectors that can be evaluated across model families.

(Huang et al., 2025), while uncertainty-aware in-context learning uses uncertainty estimates to improve output reliability (Yang et al., 2023). Semantic entropy detects hallucinations by measuring variation in the meaning of sampled generations, showing that output-side semantic variability can provide evidence of unreliable answers (Farquhar et al., 2024).

2.2. Internal Representation Based Hallucination Detection

Recent hallucination detection methods increasingly use model-internal information. The Geometry of Truth analyzes layer-wise semantic representations and shows that truth-related information can be encoded in internal activations (Mir, 2025). SymLoc studies symbolic localization of hallucination across layers, showing that hallucination-related behavior can emerge before the final output (Lamba et al., 2025).

Other work trains hallucination detectors directly on internal representations. Deep Internal Representation Analysis uses layer-wise hidden representations for detection and proposes a benchmark for evaluating such detectors (Zhang et al., 2025). Latent steering methods, such as Truthfulness Separator Vectors, separate truthful and hallucinated generations in embedding or latent space (Park et al., 2025). These approaches show that internal states are useful for hallucination detection, but they require access to representations that may be unavailable in closed or API-only deployment settings.

2.3. Distribution Shift and Cross-Model Generalization

A separate challenge is whether hallucination detectors generalize when the model or data distribution changes. HALLUSHIFT studies distribution shifts in hallucination detection and shows that detector performance can vary across settings (Dasgupta et al., 2025). This issue matters for deployment, where a detector trained on one model family may be applied to outputs from another. This motivates testing whether output-side hallucination signals transfer across LLM families.

3. Proposed Approach

3.1. Experimental Setting

Figure 2 summarizes the proposed output-side hallucination detection pipeline. We use CommonsenseQA because it provides ground-truth multiple-choice answers, with 9.7k training samples and 1.2k validation samples. Responses are generated from three open-weight model families: Llama 3.1 8B, Mistral 7B, and Gemma2 9B.

Each CommonsenseQA example is formatted as a multiple-choice prompt containing the question and all answer options. For each question, the model assigns scores to the candidate answers and generates the highest-scoring answer a. This answer is compared against the gold answer a^* to define the binary label:

$$y = \begin{cases} 1, & a \neq a^* \\ 0, & a = a^* \end{cases}$$

where $y = 1$ indicates an incorrect answer and $y = 0$ indicates a correct answer. In this closed-answer setting, incorrect answers are treated as hallucination candidates. Before MLP training, all four output-side features are standardized using statistics computed from the training split.

3.2. Output-Side Feature Extraction

Each response is represented using four output-side features including entropy, top-2 probability margin, length-normalized log probability, and token count.

Entropy measures uncertainty in the model's output distribution:

$$H(p) = -\sum_{i=1}^{K} p_i \log(p_i)$$

where p_i is the probability assigned to the i token in the distribution and K is the number of tokens considered. Higher entropy indicates greater uncertainty.

The top-2 margin measures how strongly the model prefers its most likely answer over the second most likely option:

$$M = p_{(1)} - p_{(2)}$$

where $p_{(1)}$ and $p_{(2)}$ are the highest and second highest answer probabilities. Lower margins indicate less decisive predictions.

Length-normalized log probability measures the average probability assigned to the generated answer tokens:

$$l_{norm} = \frac{1}{T_a} \sum_{t=1}^{T_a} \log P(x_t | x_{<t}, q, a)$$

where T_a is the number of generated answer tokens, x_t is the token generated at step t, $x_{<t}$ is the previous token context, q is the question, and a is the generated answer. This normalization controls for differences in response length.

The final feature vector is:

$$X = [H(p), M, l_{norm}, T_a]$$

where T_a is included as the token-count feature.

3.3. Hallucination Detector

We train an MLP to predict whether a response is incorrect from its output-side feature vector. The MLP has two hidden layers with 128 and 64 units, ReLU activations, and dropout of 0.1. The detector is trained with Adam using weighted binary cross-entropy with logits to account for class imbalance:

$$L = -\frac{1}{N} \sum_{n=1}^{N} [w_1 y_n \log(\sigma(z_n)) + w_0 (1 - y_n) \log(1 - \sigma(z_n))]$$

where N is the number of training examples, z_n is the detector logit, y_n is the binary label, and w_1 and w_0 are class weights for incorrect and correct responses.

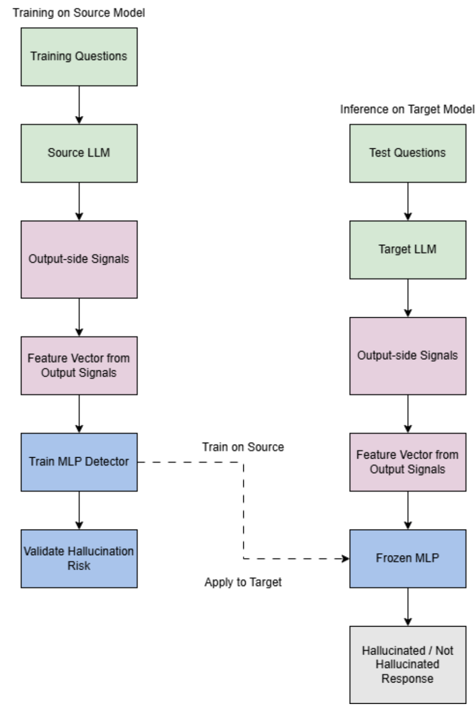

Figure 2. Overview of the proposed output-side hallucination detection pipeline. Each model answers the same CommonsenseQA questions, responses are labeled using ground truth answers, output-side features are extracted from generation, and an MLP detector is evaluated under same-model, cross-model, and pooled-training settings.

3.4. Evaluation Protocol

The detector is evaluated by same-model, cross-model, and pooled training. Same-model evaluation trains and tests on outputs from the same model family. Cross-model evaluation trains on a source model and tests on a different target model. Pooled training combines outputs from multiple source models and evaluates on a held-out target model.

We compare the MLP against single-feature baselines using entropy, top-2 margin, length-normalized log probability, or token count alone. We report AUROC as the primary metric because it measures separation between correct and incorrect responses without depending on a fixed classification threshold. Accuracy and F1-score are also reported, but as secondary metrics to show threshold-level behavior. Since correct and incorrect classes are imbalanced, a fixed threshold may not transfer cleanly across model families.

Table 1. Same-model AUROC for single-feature baselines and the multi-signal MLP.

Method	Llama	Mistral	Gemma
Entropy	55.93	65.17	59.08
Top-2 margin	53.72	65.11	58.81
Length-norm log prob.	54.85	60.62	55.79
Token count	57.80	61.73	62.49
Best single feature	57.80	65.17	62.49
Multi-signal MLP	**62.51**	**77.84**	**76.71**
Improvement	+4.71	+12.67	+14.22

4. Experiments and Results

4.1. Experimental Setup

All experiments use the dataset, labels, features, and detector described in Section 3. The results focus on AUROC as the main metric, while accuracy and F1-score are included for additional threshold-level context.

4.2. Single-Feature Baselines

Table 1 shows the same-model performance of the single-feature logistic regression baselines. These baselines test whether any individual output-side signal is strong enough to detect incorrect answers on its own.

The results show that no single feature is consistently strongest across all model families. Token count performs best for Llama and Gemma, reaching 57.80 and 62.49 AUROC, while entropy performs best for Mistral at 65.17 AUROC.

This variation suggests that different model families expose different output-side signals when they answer correctly or incorrectly. Entropy and margin are more useful for Mistral, while token count is more useful for Llama and Gemma. However, the single-feature results remain limited overall, and none of the individual features reaches the same-model performance of the MLP on any model family.

4.3. Same-Model Detection

The multi-signal MLP improves over all single-feature baselines in the same-model setting. It reaches 62.51 AUROC on Llama, 77.84 AUROC on Mistral, and 76.71 AUROC on Gemma. Compared with the best single-feature baseline, the MLP improves by 4.71 AUROC points on Llama, 12.67 points on Mistral, and 14.22 points on Gemma.

The improvement is smallest for Llama, where all methods remain relatively close to chance compared with the other model families. The larger gains on Mistral and Gemma suggest that the output-side features are more informative when used together than when used independently. This supports the use of a multi-feature detector rather than a detector based on a single uncertainty or confidence signal.

These same-model results also show that hallucination detection can be performed without model-internal activations. The detector only uses values available from the generated output and token-level generation behavior. This is important because output-side detectors are easier to reuse in settings where hidden states, intermediate activations, or model-specific internals are unavailable.

4.4. Cross-Model Transfer

Table 2 reports the main cross-model transfer results using accuracy, F1-score, and AUROC. The diagonal entries correspond to same-model evaluation, while the off-diagonal entries show transfer from one model family to another. This setting tests whether the detector learns output-side hallucination patterns that are specific to the source model, or whether some of those patterns generalize across model families. AUROC is the main comparison metric, while accuracy and F1-score provide secondary threshold-level context under class imbalance.

The results show meaningful cross-model transfer, but the strength of transfer depends on the source and target models. A detector trained on Llama reaches 65.98 AUROC on Mistral and 70.58 AUROC on Gemma. This is higher than its same-model result on Llama, suggesting that Llama-trained features can still provide useful separation on other target models. A detector trained on Mistral transfers especially well to Gemma, reaching 74.83 AUROC. Similarly, the Gemma-trained detector reaches 73.95 AUROC on Mistral. These two results suggest that Mistral and Gemma share stronger output-side correctness signals with each other than either does with Llama.

Transfer to Llama is weaker. The Mistral-trained detector reaches 60.54 AUROC on Llama, while the Gemma-trained detector reaches 61.86 AUROC on Llama. These values are above random chance, but lower than the Mistral-Gemma transfer results. Differences in calibration, tokenization, response length, and answer distributions may affect how well output-side features transfer from one model family to another. This pattern is also consistent with the answer distribution, since Llama had the strongest class imbalance among the evaluated models.

The combined-training detector gives stable performance across target models. It reaches 62.19 AUROC on Llama, 77.32 AUROC on Mistral, and 76.03 AUROC on Gemma. These results are close to the best same-model results for each target model, although combined training does not outperform the target-specific detector in every case. This suggests that pooled training can produce a general detector that remains competitive across model families. The accu-

Table 2. Multi-signal MLP results across same-model, cross-model, and combined-training settings. Values are percentages.

Metric	Training setup	Test Llama	Test Mistral	Test Gemma
Accuracy	Llama-trained	56.27	67.97	48.89
	Mistral-trained	73.62	72.56	67.81
	Gemma-trained	74.11	73.05	68.05
	Combined-trained	73.62	73.05	68.46
F1	Llama-trained	64.59	48.48	33.67
	Mistral-trained	84.81	63.39	76.09
	Gemma-trained	84.81	47.98	69.69
	Combined-trained	84.47	49.54	71.34
AUROC	Llama-trained	**62.51**	65.98	70.58
	Mistral-trained	60.54	**77.84**	74.83
	Gemma-trained	61.86	73.95	**76.71**
	Combined-trained	62.19	77.32	76.03

racy and F1-score values follow a less consistent pattern than AUROC, which supports treating them as secondary metrics rather than the main basis for comparing transfer performance.

Overall, the results support the main hypothesis of this paper. Output-side signals contain useful information for detecting incorrect generations. The same-model results show that combining multiple output-side features improves performance over single-feature baselines, and the cross-model AUROC results show that the learned detector can transfer across LLM families, especially between Mistral and Gemma.

5. Discussion

The results show that output-side signals can provide useful information for hallucination detection across different LLM families. In the same-model setting, the multi-signal MLP consistently outperforms the single-feature baselines. This suggests that no individual signal fully captures whether a model answer is likely to hallucinate. Instead, the detector benefits from combining uncertainty, confidence margin, length-normalized probability, and response length into a shared feature representation.

The cross-model results further suggest that hallucination-related output patterns are not entirely model specific. Detectors trained on one model family often retain useful performance when evaluated on another model family. These results support that output-side detectors may be easier to reuse across model families than methods that depend on hidden states or internal activations.

However, pooled training does not clearly outperform the target-specific detector. This suggests that combining model outputs can make the detector more general, but adding more source models does not automatically improve performance on every target. Transfer to Llama is also weaker than transfer between Mistral and Gemma, which suggests that calibration, tokenization, response length, answer formatting, and label imbalance can affect how well output-side features transfer.

There are two limitations to this study. First, the task uses CommonsenseQA, where each question has a fixed multiple-choice answer. This makes labeling straightforward, but it also simplifies the hallucination detection problem. In this setting, an incorrect answer is treated as a possible hallucination, even though not every wrong answer reflects the same kind of hallucination. Some errors may come from ambiguity, weak commonsense reasoning, or answer extraction issues rather than fabricated information.

Second, the evaluation does not test free-form generation. Open-ended answers can contain partially correct statements, unsupported claims, or factual errors that are harder to label with a single binary correctness value. Since this study uses closed-answer evaluation, the results may not fully represent hallucination behavior in longer generations, summarization, dialogue, or open-domain question answering.

6. Conclusion and Future Work

This paper studied whether hallucination detection can be performed using output-side signals that transfer across different LLM families. Instead of relying on model-internal representations, the proposed detector uses output signals. These signals are easier to access in deployment settings where hidden states, activations, or latent representations may not be available.

The results show that combining multiple output-side signals with an MLP improves over single-feature baselines in the same-model setting. The detector also shows meaningful cross-model transfer, especially between Mistral and Gemma. These findings suggest that some hallucination-related output patterns are shared across model families, although transfer remains uneven across targets.

Future work should extend this approach beyond closed-answer CommonsenseQA evaluation. Open-ended question answering would provide a more realistic test because generated responses can be partially correct, unsupported, or factually wrong in different ways. Another direction is to test whether output signatures from larger teacher models can help train detectors for smaller models.

7. Team Contribution Statement

Project presentation slide preparation, code development, demo preparation, and final report writing were done by Santiago Posse.

8. Use of External Code and AI Tools

Open AI's Codex was used during the code development for creating the boilerplate function for the LLM 4-bit quantization loading. Codex was additionally used for debugging errors.

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